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Variable Natural Frequency Damper for Minimizing Response of Offshore Wind Turbine: Principle Verification through Analysis of Controllable Natural Frequencies

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Abstract: Resonance causes extreme stress, acceleration of fatigue, and reduction in lifespan of offshore wind structures. The main factors that cause resonance are environmental loads such as wind and waves, and dynamic loads caused by rotor movement. Estimation of the natural frequency at the design stage is highly uncertain, and natural frequency changes occur due to various factors during long-term operation. Therefore, it is important to ensure structural safety from resonance through a vibration-monitoring system or an additional damper. In this study, the effect of seawater existing inside the substructure on the natural frequency of the structure was dealt with. The natural frequency estimation equation for a fixed offshore wind structure was derived with the “inner fluid simplification assumption”. The finite element modal analysis was performed to verify the principle of Variable Natural Frequency Damper (VNFD), a system that controls the natural frequency of offshore wind structures through a pump, and to find the range of natural frequency control. As a result, interior fluid affects the natural frequency of the wind turbine support structure. Specifically, the variable natural frequency range was very low, at about 0.027% for the monopile model at a depth of 10 m, but increased rapidly to about 3.66% at a depth of 70 m. Furthermore, when estimating the natural frequency of a fixed offshore wind turbine in deep water without consideration of interior fluid, the estimates can be higher than with consideration of it.

Keywords: offshore wind turbine; substructure; damper; natural frequency; fluid structure interaction; soil structure interaction; linear modal analysis



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1. Introduction

Installation of wind turbines has increased significantly around the world over the past few years, and special attention is being paid to offshore wind power installations, which are relatively advantageous in terms of efficiency and development location compared to onshore wind power.

Generally, the offshore wind substructure is designed by the Load and Resistance Factor Design (LRFD) method [1] according to the international design standards [2] for the Ultimate Limit State (ULS) and Fatigue Limit State (FLS) of the working environment. ULS corresponds to the resistance of maximum load, such as typhoons and earthquakes, while FLS refers to the accumulation of fatigue in structures due to repeated loads (stresses) such as blade dynamic loads, wind, and waves.

Offshore wind turbines are being installed at sites deeper and farther from the coast, and the turbine systems supported by substructures are becoming increasingly large [3]. Accordingly, the substructure supporting the turbine system also tends to be bigger. Substructures are classified into various types, but monopile and jacket types are the most popular in Europe, and their installed share in 2018 was 81.7% and 6.9%, respectively [4]. In general, the substructure to be used is selected depending on the depth of water at the

installation site, the condition of the seabed, and the model of the turbine to be installed. The resonance phenomenon of the structure, due to other loads such as cyclic loading, should be avoided, and special attention should be paid to the determination of the stress range and the amount of damping of the excited structure in the resonance range during fatigue design [5].

Resonance caused by wind and waves accumulates fatigue in structures and reduces their lifespan. However, resonance still occurs due to various conditions that are difficult to predict during the design stage, such as harmonic-type winds and waves and their misalignment, and break waves [6]. In addition, changes in soil structural stiffness and response due to cyclic lateral loading can lead to structural resonance and risk of fatigue damage [7].

When designing a fixed offshore wind turbine structure, it is common to design a soft-stiff so that the natural frequency of the structure is between the turbine rotation speed (1P) and the blade passing frequency (3P). The monopile type is not suitable for deep water and is known to be economical at depths of 30 m or less. In particular, in the case of dynamic characteristics, as the capacity of the turbine increases, the natural frequency of the structure becomes closer to the 1P frequency caused by the rotor rotation due to the long and slender substructure. To avoid this, the transition piece connecting the turbine system and the monopile must be designed stiffly. In addition, turbine models with a wide rotor rotational speed range (4.5–14.8 rpm), such as Areva M5000 5 MW, does not have a soft-stiff region, so a sophisticated pitch-control system is designed. Resonance is avoided by skipping through this pitch control, but this reduces economic feasibility [8].

Therefore, in order to solve these problems, this paper focuses on the study of VNFD, a system that is designed to actively avoid the resonance phenomenon. VNFD avoids resonance by controlling the Inner Water Level (IWL) of the substructure with the seawater pump when the frequency of the external force and the natural frequency of the structure match.

The systematic logic of VNFD is shown in Figure 1a and the theoretical configuration is shown in Figure 1b. The VNFD system is designed such that it compares the natural frequency ($f_{natural}$) with the forcing frequency (f_{forced}) and, whenever the natural frequency is lower than the forcing frequency, seawater is supplied to the inside of the substructure and, whenever the natural frequency is greater than the forcing frequency, the seawater inside the substructure is drained out. The process of calculating natural frequency and forcing frequency can be achieved using the SCADA system, but other methods can also be used in the actual application because it is difficult to post-process the vibration data every time it is required. The inside of the monopile corresponding to the height of the seabed can be assumed to be filled with soil, so it is assumed that the bed equal to the height of the seabed is also present inside the substructure. Therefore, the controllable IWL range is from the seabed to the platform height. Generally, the inside of the transition piece contains some seawater due to the diameter gap of the two tubular steels when grouting. However, sea water may infiltrate by seal leakage related to the bottom outlet of the substructure wall and J-tube and, as a result, tidal changes may occur inside the foundation [9]. Therefore, unless additional measures are taken, an internal coating of steel may be required for the intended operation of the VNFD, and treatment for sealing may be required. Also, since hydrostatic pressure is applied to the tubular steel as much as the difference between the external water level and the inner water level, this must be considered when applying this technology.

In this paper, after introducing the simplified formula derivation for predicting the natural frequency of the monopile type structure with the seawater inside the substructure, three-dimensional Finite Element (FE) modal analysis was performed using ANSYS 2021R1 to find the range of change in the natural frequency of offshore wind structures according to IWL. According to Arany et al. [8], measured and predicted natural frequencies of wind turbine structures have been reported in various studies [10–15], and the causes of these differences are either the theory or physical idealization of the system or parameters, or

a combination of the two. However, the error caused by ignoring the water inside the substructure was not considered. Since the natural frequency of a wind turbine structure is affected by mass, the mass must be considered for accurate prediction, which is particularly important in monopile types with large internal water or deep-water models. Infiltration of seawater and other outlets as described above were not considered. In Section 2, the latest research related to natural frequencies of offshore wind structures is investigated, and theories related to simulation are introduced. Section 3 describes the FE model, and the results and conclusions are discussed.

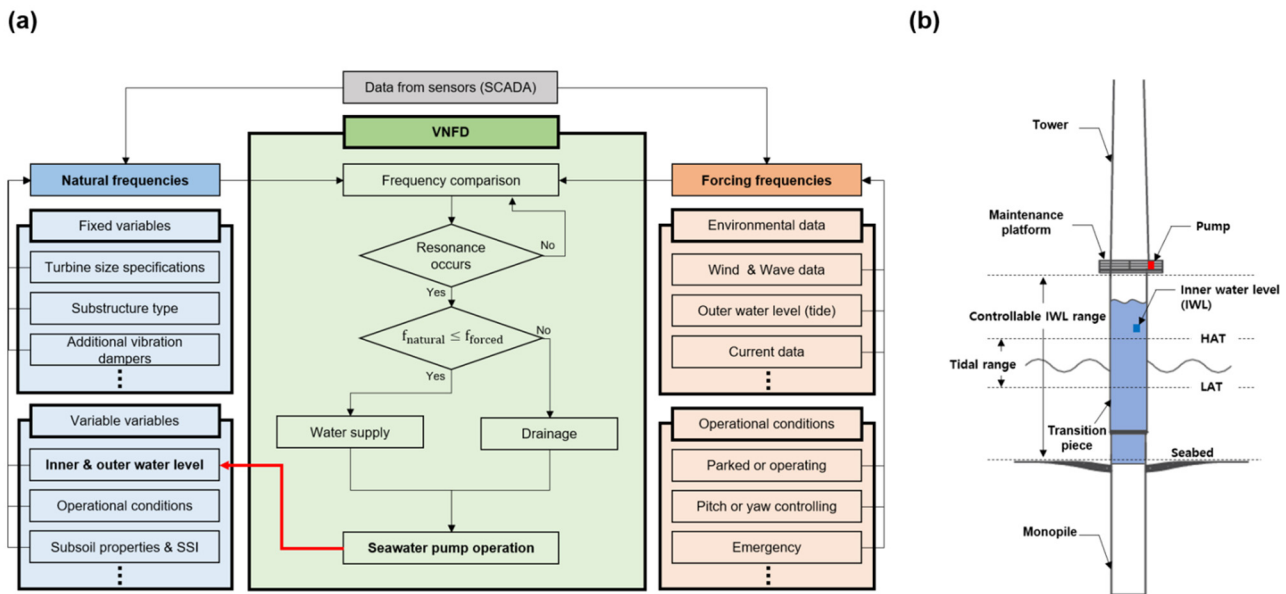


Figure 1. (a) System logic of VNFD and (b) theoretical configuration.

2. Materials and Methods

2.1. Natural Frequency of Offshore Wind Turbine Structure with Consideration of Inner Water

Modal analysis is used in determining the vibration characteristics of structures, and vibration characteristics such as natural frequency and mode shape of structures are important in the design of offshore wind structures that are regularly subjected to dynamic loads. In terms of the element equations of the FE modal analysis, damping and external forces are ignored and only the inertia and stiffness terms remain. The basic equation for the general eigenvalue problem in undamped modal analysis is Equation (1).

$$k\varphi_i = \omega_n^2 m\varphi_i \quad (1)$$

where, φ_i is the mode shape vector for mode i , and ω_n is the natural frequency for mode. Various studies have been conducted on the natural frequency theory of offshore wind structures [16,17]. The first natural frequency for a single degree of freedom (SODF)-concentrated mass model of a wind turbine system with a constant tower cross-section can be expressed as Equation (2).

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{k_T}{m_{RNA} + \alpha m_T}} \quad (2)$$

where, $k_T = \frac{3E_T I_T}{L_T^3}$ means the lateral stiffness of the tower, and E_T , I_T , and L_T are the tower's Young's modulus, moment of inertia, and length, respectively. Furthermore, m_{RNA} is the mass of rotor-nacelle assembly (RNA), m_T is the mass of the tower, and α is the equivalent mass ratio of the tower concentrated to the tower top. According to van der Tempel and Molenaar [18], when the wind turbine is considered as an inverted pendulum

with flexural stiffness ($E_T I_T$) and tower mass per meter (m_t), the first vibrational mode is equal to Equation (3):

$$f_1 \cong \sqrt{\frac{3.04 E_T I_T}{(m_{RNA} + 0.227 m_t L_T) 4 \pi^2 L_T^3}} \quad (3)$$

However, most circular wind turbine towers have a tapered shape in which the diameter and thickness of the cross-section become smaller as the altitude increases, and many simplified methods have been used [18–20]. According to Yung-Yen Ko [20], the closed-form solution of the natural frequency of wind turbine structures with linearly tapered towers in a rigid-base condition is given by Equation (4):

$$f_{RB(tapered)} = \frac{1}{2\pi} \sqrt{\frac{k_{T(tapered)}}{m_{RNA} + \alpha m_{T(tapered)}}} \quad (4)$$

where, $k_{T(tapered)}$ is the lateral stiffness of a tapered tower [8], and $m_{T(tapered)}$ is the mass of the tapered tower. However, in Equation (4), only the turbine system starting from the tower base was considered. When the substructure is taken into account, the structure becomes more flexible because it has an additional length with a length proportional to the water depth. Utilizing Castigliano's second theorem, equivalent lateral stiffnesses can be derived assuming that the section rigidity of the monopile continues to the bottom of the tower [8]. The natural frequency of the structure considering the length of the substructure can be obtained through Equation (5) [20]:

$$f_{RB(tapered+SS)} = \frac{1}{2\pi} \sqrt{\frac{k_{T(tapered+SS)}}{m_{RNA} + \alpha m_{T(tapered+SS)}}} \quad (5)$$

where, SS is an abbreviation of substructure, and $k_{T(tapered+SS)}$ and $m_{T(tapered+SS)}$ are the lateral stiffness and mass of the tapered tower considering the substructure, respectively.

Inside the monopile substructure, sea water flows in when the foundation is installed as shown in Figure 1b, and its weight is calculated as in Equation (6).

$$m_w = \frac{1}{4} \pi \rho_w H (D_P - 2t_P)^2 \kappa \quad (6)$$

where ρ_w , H , D_P , and t_P are water density, depth, monopile diameter, and thickness, respectively. Lastly, κ is the ratio of water in the interior to the water depth, which is 1.0 when the inner water level is the same as the water depth. For the first natural frequency mode of the cantilever structure, the distance away from the support and the mass value contribute to the vibration mode. Since m_w has a different distance from the mudline than RNA, it is desirable to consider it along with the equivalent mass ratio (α_w), and the following "inner fluid simplification assumption" is necessary.

- (i) When the dynamic response of the structure is transmitted to the inner water, a sloshing phenomenon occurs due to the inertial effect of the fluid, but this is ignored for the sake of simplification and only considered as a mass.
- (ii) Since m_w is not a mass supported by the structure, it is complicated to calculate the mode participation for it. Therefore, the empirical factor (ψ) was applied.

Therefore, the equivalent mass ratio (α_w) can be assumed to be the mass ratio to the distance of center of gravity with ψ as in Equation (7):

$$\alpha_w = \psi \frac{m_w (CG_w + H)}{m_{RNA} (CG_{RNA} + H)} = \psi \frac{0.5 m_w H \kappa}{m_{RNA} (L_T + H)} \quad (7)$$

where the center of gravity of the RNA (CG_{RNA}) is the same as L_T because the sea level is the standard of the height, and the center of gravity of the inner water (CG_w) can be calculated as $0.5H(\kappa - 2)$. To consider the distance from the mudline, the depth (H) must

be added to CG_{RNA} and CG_w , respectively. Therefore, the natural frequency considering the additional mass of inner water along with “inner fluid simplification assumption” and can be expressed as Equation (8).

$$f_{RB(tapered+SS+w)} = \frac{1}{2\pi} \sqrt{\frac{k_{T(tapered+SS)}}{m_{RNA} + \alpha m_{T(tapered+SS)} + \alpha_w m_w}} \quad (8)$$

There are many studies on the interaction of soil and piles for eigenvalue problems of offshore wind structure [8,16,20–22]. Using the simplified soil-structure interaction (SSI) analysis model proposed in Ref. [23], the flexibility of substructure and foundation can be considered simultaneously with lateral stiffness (k_h) and rotational stiffness (k_r), which is given by Equation (9):

$$f_{FF(tapered+SS+w)} = \frac{f_{RB(tapered+SS+w)}}{\sqrt{1 + k_T/k_h + k_T L_T^2/k_r}} \quad (9)$$

To analyze these resonant frequencies, the log data of the acceleration sensor through the CMS system or SCADA system must be processed using valid data. It is necessary to read the index value of the acceleration sensor and judge the situation while considering not only the operation status of the wind turbine such as start-up, stop, pitching, and yawing, but also wind and wave data; however, this is a lot of work and it is not organized. In particular, it is not easy to establish the effect of the dynamic loads of the rotor and blades with valid data [7]. For this reason, accurate natural frequency estimation is very difficult and unreliable as described above. The natural frequency of an offshore wind turbine always changes depending on the surrounding environment and turbine operation conditions, and it is very difficult to predict at the design stage because different damping factors appear depending on these conditions.

As a result of analyzing the acceleration sensor data for two weeks of an offshore wind power generator installed in the North Sea of Belgium, it was confirmed that the structural frequency changes due to the tidal difference [7]. Through this, it can be seen that the external water level affects the vibration frequency of the structure. Moreover, according to one experiment, the natural frequency for the first mode of the tank containing the fluid decreased as the amount of fluid increased [24]. Therefore, it can be assumed that not only the external fluid but also the IWL affect the natural frequency of the structure, and it is expected that the higher the IWL, the lower the natural frequency of the structure.

When designing an offshore wind structure, it is recommended to avoid the dynamic load frequency due to the rotational speed of the blade and rotor. Accordingly, DNVGL-ST-0126 [5] suggests a 5% range as in Equation (10) so that the excitation frequency and natural frequency of the structure differ by more than 5%.

$$\frac{f_R}{f_{0,n}} \leq 0.95 \text{ or } \frac{f_R}{f_{0,n}} \geq 1.05 \quad (10)$$

where f_R is the excitation frequency, especially the rotational frequency range of the rotor and the transition frequency of the rotor blades in the normal operating range, $f_{0,n}$ is the n th natural frequency of the tower and the entire system of the foundation. Additionally, it is recommended to have a margin of $\pm 5\%$ in consideration of the imbalance of the soil and the uncertainty of using the software. It is said that this requirement can be omitted if measures are in place to prevent resonance effects, such as vibration monitoring systems or damping devices [5]. Therefore, international design standards also recommend monitoring based on experience, and an empirical margin far exceeding the avoidance range is recommended.

Since VNFD is different from a general damper in principle, how much influence it will have on these standards can be known through research considering the sloshing effect of inner water and applying it together with other dampers.

2.2. Linear Fluid-Structure Interaction Using ANSYS Code

For the dynamic problem of calculating the natural frequency through the FE model, the modal analysis enables a direct approach. To use this method, eigenvectors must be calculated by solving the symmetric eigenvalue problem, and various formulas can be used for fluid-structure coupled analyses [25].

The three-dimensional Computational Fluid Dynamics (CFD) numerical method is advantageous because the surface pressure distribution can be obtained through CFD software, and this pressure is applied to the structural system. According to [26], how the interactions between fluids and solids are described offers the greatest opportunity of reducing the computational effort, and by using one-way simulations where only fluid pressure data are applied to structures, the computational effort of structural simulation can be greatly reduced.

The equations for coupled fluid-structure problems using pressure-based equations for fluids are described in the frequency domain. The structure problem is written as Equation (11), and the fluid problem is written as Equation (12):

$$-\omega^2 \int_{\Omega_S} \rho_S u_i \delta u_i - \int_{\Gamma} p n_i \delta u_i = 0 \quad (11)$$

where Ω_S is the structure domain with boundary: $\partial\Omega_S = \partial\Omega_{S\sigma} \cup \partial\Omega_{SO} \cup \Gamma$ where $\partial\Omega_{S\sigma}$ is the boundary part with imposed forces, $\partial\Omega_{SO}$ is the boundary part with imposed displacement, and Γ is the fluid-structure interface. Furthermore, u is the structure displacement, ρ_S stands for structure density, $\mathbf{n}$ is the inward normal on Γ , $\mathbf{n}_S$ is the outward normal on $\partial\Omega_S$ and $\sigma(\mathbf{u})$ is the stress tensor.

$$\int_{\Omega_F} \frac{\partial p}{\partial x_i} \frac{\partial \delta p}{\partial x_i} - \omega^2 \int_{\Omega_F} \frac{p \delta p}{c^2} - \rho_F \omega^2 \int_{\Gamma} u_i n_i \delta p = 0 \quad (12)$$

where Ω_F is the fluid domain with $\partial\Omega_F = \partial\Omega_{FO} \cup \partial\Omega_{SO} \cup \Gamma$, where $\partial\Omega_{F\pi}$ is the boundary part with imposed normal gradient pressure (rigid wall or symmetry plane), and $\partial\Omega_{FO}$ the boundary part with imposed pressure (pressure release surface or anti-symmetry plane). The vector $\mathbf{n}$ is the outward normal on Γ , and $\mathbf{n}_F$ is the outward normal on $\partial\Omega_F$. The fluid density and velocity waves are denoted ρ_F and c respectively.

For spatial discretization using finite elements, the mass and stiffness matrices for fluid and structural problems are defined by Equations (13)–(16). The fluid structure interaction matrix used to model the coupling term is defined as Equations (17) and (18):

$$\int_{\Omega_S} \rho_S u_i \delta u_i \rightarrow \delta \mathbf{U}^T \mathbf{M}_S \mathbf{U} \quad (13)$$

$$\int_{\Omega_S} \sigma_{ij}(\mathbf{u}) \epsilon_{ij}(\delta \mathbf{u}) \rightarrow \delta \mathbf{U}^T \mathbf{K}_S \mathbf{U} \quad (14)$$

$$\int_{\Omega_F} \frac{\partial p}{\partial x_i} \frac{\partial \delta p}{\partial x_i} \rightarrow \delta \mathbf{P}^T \mathbf{K}_F \mathbf{P} \quad (15)$$

$$\int_{\Omega_F} \frac{p \delta p}{c^2} \rightarrow \delta \mathbf{P}^T \mathbf{M}_F \mathbf{P} \quad (16)$$

$$\int_{\Gamma} p n_i \delta u_i d\Gamma \rightarrow \delta \mathbf{U}^T \mathbf{R} \mathbf{P} \quad (17)$$

$$\int_{\Gamma} u_i n_i \delta p d\Gamma \rightarrow \delta \mathbf{P}^T \mathbf{R}^T \mathbf{U} \quad (18)$$

Finally, the coupled problem is defined in Equation (19):

$$\begin{bmatrix} \mathbf{K}_S & -\mathbf{R} \\ 0 & \mathbf{K}_F \end{bmatrix} \begin{Bmatrix} \mathbf{U}(\omega) \\ \mathbf{P}(\omega) \end{Bmatrix} = \omega^2 \begin{bmatrix} \mathbf{M}_S & 0 \\ \rho_F \mathbf{R}^T & \mathbf{M}_F \end{bmatrix} \begin{Bmatrix} \mathbf{U}(\omega) \\ \mathbf{P}(\omega) \end{Bmatrix} \quad (19)$$

where $U(\omega)$ and $P(\omega)$ are the displacement and pressure in frequency domain, respectively, and R is fluid-structure interaction matrix. The mass and stiffness of structure and fluid are denoted M_S , K_S , M_F , and K_F , respectively. The displacement–pressure formulation leads to an unsymmetric eigenvalue problem that can be solved with the unsymmetric Lanczos algorithm [27,28].

In this paper, the ANSYS modal acoustic analysis model was used for modal analysis considering fluid–structure interaction (FSI). The interaction between the structure and the fluid is formed by pressure rather than velocity [29]. This involves modeling the fluid medium as well as the surrounding structure to determine frequencies and standing wave patterns within the structure. Instead of using the two-way FSI analysis in which the deformation of the structure is reflected in the fluid solver at the same time, by using the one-way FSI, the VNFD can reduce damping caused by the air-conditioning of the fluid inside the substructure such as the principle of other additional dampers. Ignoring it means accepting the interaction with the fluid outside the substructure in the form of one-way pressure, and through this assumption, the efficiency of the analysis can be greatly improved.

2.3. Three-Dimensional Soil-Structure Interaction

The Soil–Structure Interaction (SSI) is a very important factor in vibration studies because it is the largest damping factor in offshore wind turbine structures [30,31], as well as designing capacity [9,32]. Jalbi et al. [33] broadly classified methods to consider these SSI into three groups as simplified SSI, standard SSI (see Figure 2a), and advanced method (finite element shown in Figure 2b). In this paper, the FE model method is used because it is difficult to converge the analysis using p-y, t-z, and q-z spring [34], which is a nonlinear spring theory, and it is convenient to implement the condition in which the fluid elements inside the substructure is supported by soil elements. Additionally, the layered soil in this paper consist of sand. In general, unlike sand, clay exhibits physical properties such as a high damping factor due to its adhesion. If clay is used instead of sand, and other dynamic characteristics and other conditions are the same, the natural frequency is generally reduced.

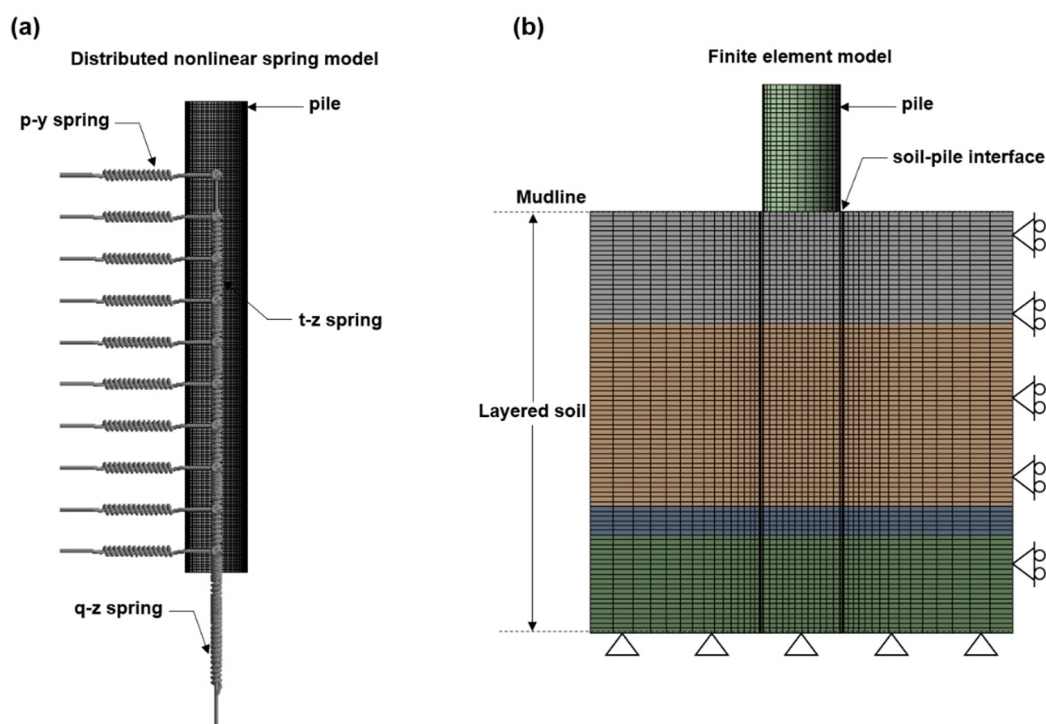


Figure 2. Soil-pile interaction of (a) distributed nonlinear spring model and (b) FE model.

Figure 3. Dimensions of (a) NREL 5 MW and OC3 monopile embedded in layered sandy soil except for water depth and (b) detail of grouted connection.

Table 2. Material properties of steel, grout [36].

Material	Young's Modulus (GPa)	Poisson's Ratio (-)	Density (kg/m ³)
Steel	210	0.38	8500
Grout	70	0.19	2740

Table 3. Material properties of soil layers [38].

Material	Depth (m)	γ (kN/m ³)	c (kPa)	ϕ (Degree)	ν (-)	E_s (MPa)	Dilation Angle (Degree)
Sand 1	0.0–5.0	10.0	0.05	33.0	0.3	30.0	3.0
Sand 2	5.0–14.0	10.0	0.05	35.0	0.3	35.0	5.0
Sand 3	14.0–36.0	10.0	0.05	38.5	0.3	47.0	8.5

For the contact between the grout and the steel interface, a bonded contact that does not allow sliding or gaps was used. Bonded contact conditions in three-dimensional FE analysis permanently hold the target and contact surfaces [39–41]. The frictional coefficient of contact between the soil and the structure is obtained through the angle of friction (ϕ) in Table 3 [38]. However, through the linear assumption of the modal analysis, this non-linearity is regarded as a bonded contact condition. In the case of water, only pressure is transmitted, and it is assumed that there is no gap and no lateral friction, so the no separation option was set [42]. The contact elements of linearly simplified contact conditions for linear modal analysis were modeled with CONTA174 and TARGE170. The elements of soil and structure were modeled as 20-node SOLID186, and the inner fluid for the coupled acoustic structural interaction was modeled as 3D 20-node acoustic solid, FLUID220.

For the boundary condition, the outer and bottom surfaces of the soil were set to be completely fixed except for vertical DOF of outer surfaces (see Figure 2b), and the gravitational acceleration was 9.81 m/s² in the -z direction.

3.2. Verification of FE Models

To secure the reliability of the meshing method of the FE model, a convergence test of the natural frequency analysis results was performed while increasing the number of elements using the soil lateral direction, height direction, and bias as parameters, and the results are shown in Table 4. The difference term in Table 4 is the error rate for the natural frequency of each test step and the natural frequency of the previous step. The reason for applying the parameter based on the soil is that, the discretization of the lateral, axial, and circumferential directions of the soil coincides with the nodes at the interface of the elements of the structure or fluid. This technique is also used in many other studies using FE analysis, and the reliability of the FE model can be secured [36,43,44]. The soil model used in this process used a different model from the scenarios of this paper, but other conditions were the same, so it was judged that the reliability test result of the meshing method could be accepted.

Table 4. Results of mesh convergence test.

Test No.	Number of Elements (-)	Natural Frequency (Hz)	Difference (%)
1	5148	0.26025	-
2	6894	0.26030	0.0192
3	8580	0.26046	0.0615
4	10,386	0.26055	0.0346
5	13,788	0.26064	0.0345
6	18,673	0.26073	0.0345
7	24,275	0.26072	−0.0038
8	35,700	0.26077	0.0192

Eight mesh convergence tests were performed, and small differences occurred from test 6 as shown in Figure 4. As a result, the soil elements were divided into 32 pieces in the circumferential direction and 1500 mm in the axial direction, and 8 layers were gradually densely formed in the area adjacent to the structure in the transverse direction. The diameter of the soil model was modeled as 20 times the monopile diameter (see Figure 5).

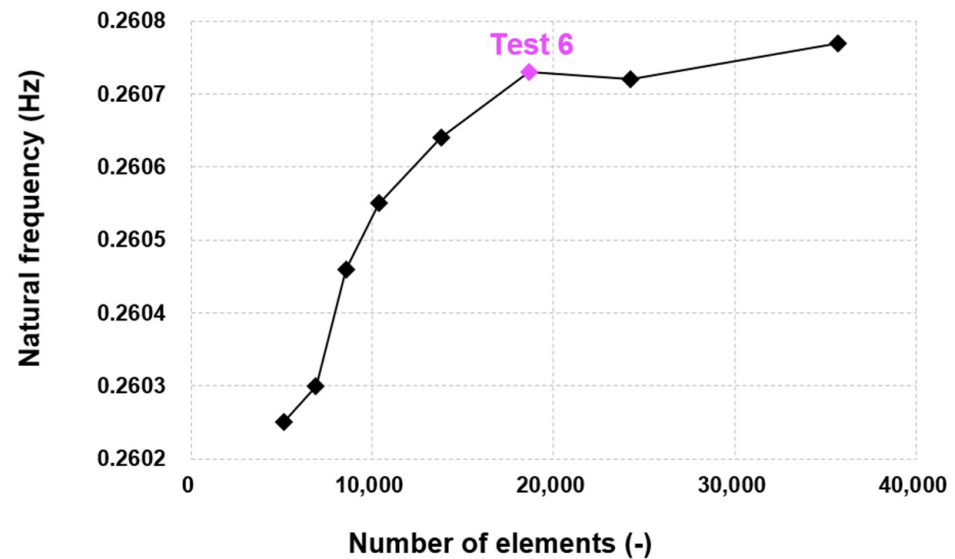


Figure 4. Mesh convergence graph.

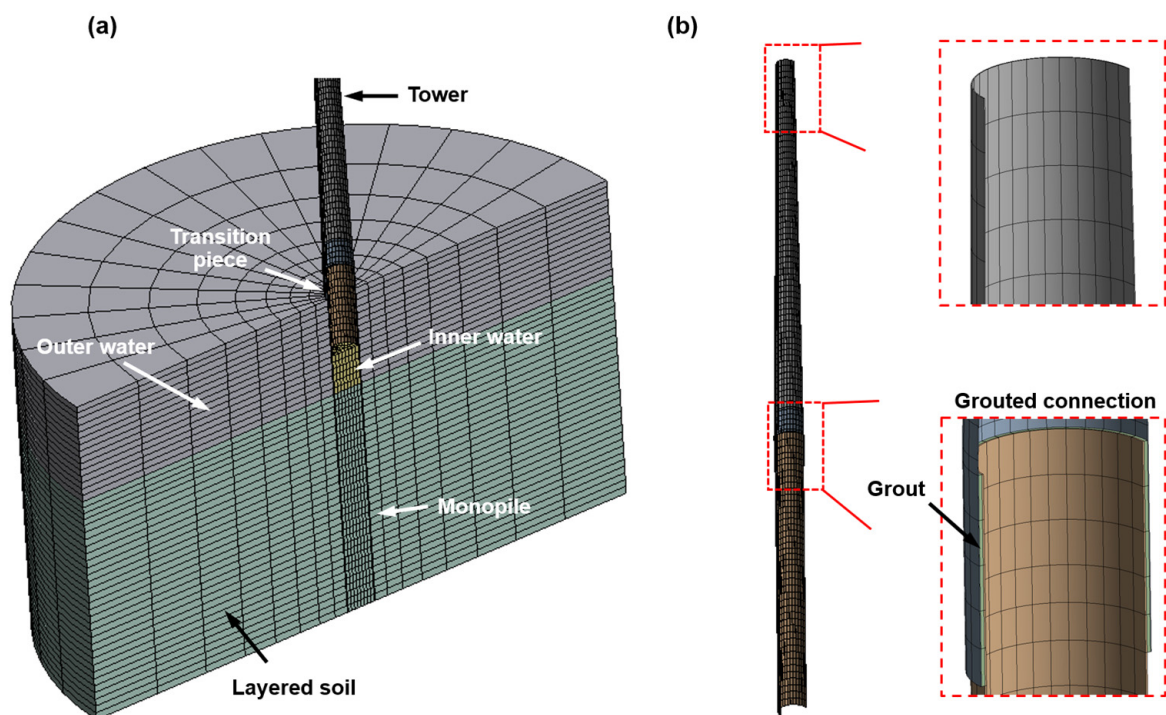


Figure 5. FE model for (a) structures with soil layers and (b) tubular steels.

In order to secure the reliability of the tower model, as shown in Table 5, the simulation results of the FE model of this paper were compared with the results of other studies of the NREL turbine model. Since the main concern of VNFD is the soft–soft domain or soft–stiff domain, it is the first natural frequency, so the results are analyzed only for the first mode. As a result, the natural frequency of the rigid base model was about 0–4% lower than that of other studies, considering only the shape of the tower. The result of fixed soil in which

the fixed condition was set in the mudline was about 0–0.5% higher. Considering the error of the rigid base model, it is assumed that the error in the geometry of the sub-structure and the error in the tower are offset. Through the results of the flexible soil model, it is expected that an error of about 2–5% occurs from the physical properties of the soil or the interaction between the soil and the foundation. Consequently, the model for the first mode analysis and the model's meshing method are acceptable.

Table 5. Comparison of the simulation results of the FE model with the results of other studies of the NREL turbine model for the rigid base (RB), fixed soil (FS) and flexible soil (FF) condition.

Support Conditions	Method	1st Tower Fore-Aft (Hz)	1st Tower Side-to-Side (Hz)	Relative Difference (%)	
				1st Tower Fore-Aft	1st Tower Side-to-Side
Rigid base	$f_{RB, FE}(\text{present})$	0.3245	0.3250	-	-
	FAST [35]	0.3240	0.3120	0.14	4.00
	ADAMS [35]	0.3195	0.3164	1.54	2.65
	$f_{RB}[20]$	0.3123	0.3123	3.76	3.91
	$f_{RB(\text{tapered})}[20]$	0.3132	0.3132	3.48	3.63
Fixed soil	$f_{FS, FE}(\text{present})$	0.2763	0.2766	-	-
	$f_{FS}[36]$	0.2760	0.2780	0.09	-0.51
	$f_{FS}[45]$	0.2770	0.2780	-0.27	-0.51
Flexible soil	$f_{FF, FE}(\text{present})$	0.2548	0.2550	-	-
	$f_{FF}[38]$	0.2420	0.2410	5.02	5.49
	$f_{FF}[36]$	0.2450	0.2480	3.85	2.75

4. Results and Discussion

4.1. Impact of Inner Water Level and Water Depth Parameters

The theoretical manageable IWL range through VNFD is from the seabed of the installation site to the height of the maintenance platform, and is determined according to the depth of the site. A monopile type is considered to be economical when the water depth is not more than 30 m [3], but the water depth was selected as a parameter to investigate the effect of the installation depth on the variable natural frequency region. Therefore, the effect of the IWL on the natural frequency of the entire structure model, except for rotor nacelle assembly, (RNA) was analyzed for a depth of between 10 to 70 m, and the controllable natural frequency range using VNFD was analyzed. When the manageable IWL was fully drained out, it was named M0, and when it was fully supplied, it was named M100, and IWL parameters were set from 0 to 100% in 20% increments. Through this, natural frequency analysis according to the change of IWL was conducted. The name of the analysis scenarios to which parameters are applied are shown in Table 6.

Table 6. Parameter setting (water depth for site & inner water level inside of substructure).

IWL (%)	Depth (m)						
	10	20	30	40	50	60	70
0	D10M0	D20M0	D30M0	D40M0	D50M0	D60M0	D70M0
20	D10M20	D20M20	D30M20	D40M20	D50M20	D60M20	D70M20
40	D10M40	D20M40	D30M40	D40M40	D50M40	D60M40	D70M40
60	D10M60	D20M60	D30M60	D40M60	D50M60	D60M60	D70M60
80	D10M80	D20M80	D30M80	D40M80	D50M80	D60M80	D70M80
100	D10M100	D20M100	D30M100	D40M100	D50M100	D60M100	D70M100

The natural frequency results of first mode according to water depth and IWL are shown in Tables 7 and 8. Based on the rotor, the results for the fore-aft (FA) direction are shown in Table 7, and the results for the side-to-side (SS) direction are shown in Table 8. There was a difference between the FA mode and the SS mode as the center of gravity of RNA was spaced apart from the axis of the support structure, and it can be seen that the difference was not large. Therefore, we will discuss only the FA mode results from now

on. In the previous work, the natural frequency of the first mode of 5.0 MW turbine model is generally in the range of 0.115 to 0.250 Hz [46]. The first fore–aft natural frequencies of this study were analyzed in the range from 0.1520 (D70M100) to 0.2802 (D10M0) Hz from Table 7.

Table 7. Results of natural frequency analysis for first fore–aft mode.

IWL (%)	Depth (m)						
	10	20	30	40	50	60	70
0	0.27984	0.25591	0.23333	0.21222	0.19257	0.17441	0.15775
20	0.27983	0.25590	0.23331	0.21220	0.19254	0.17438	0.15771
40	0.27982	0.25587	0.23326	0.21212	0.19242	0.17421	0.15751
60	0.27981	0.25582	0.23314	0.21189	0.19205	0.17369	0.15683
80	0.27979	0.25573	0.23289	0.21139	0.19121	0.17245	0.15518
100	0.27976	0.25558	0.23246	0.21046	0.18959	0.17003	0.15196

Table 8. Results of natural frequency analysis for first side-to-side mode.

IWL (%)	Depth (m)						
	10	20	30	40	50	60	70
0	0.28018	0.25617	0.23352	0.21237	0.19267	0.17448	0.15780
20	0.28017	0.25615	0.23350	0.21234	0.19264	0.17445	0.15777
40	0.28016	0.25613	0.23345	0.21226	0.19252	0.17429	0.15756
60	0.28015	0.25608	0.23333	0.21203	0.19215	0.17376	0.15688
80	0.28013	0.25598	0.23308	0.21153	0.19131	0.17253	0.15523
100	0.28010	0.25584	0.23265	0.21059	0.18969	0.17010	0.15201

Figure 6 shows the natural frequency range by IWL change as a percentage. Overall, the deeper the water depth, the lower the natural frequency of the support structure linearly, and the lower the IWL, the higher the natural frequency nonlinearly, and this will be discussed in Section 4.2. The controllable range of natural frequency of the 10 m was about 0.027%, which was very small, but at a depth of 70 m, the relative frequency was 3.666%, which was relatively large. Since the splash zone was ignored and the external water level was considered constant, only the influence on the IWL was used as a variable.

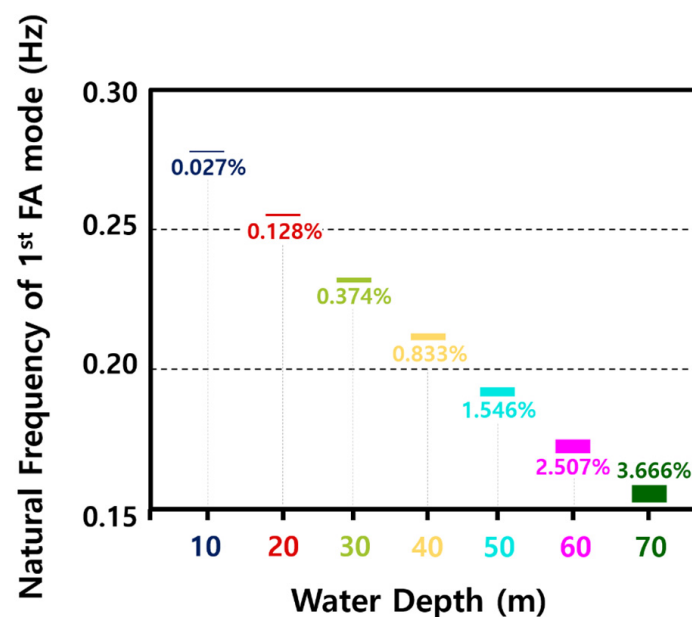


Figure 6. The variable frequency range of 1st fore–aft mode via VNFD.

Table 9 shows the comparison of the natural frequencies estimated without considering the internal fluid for the flexible soil model ($f_{FF(tapered+SS)}$) and the estimated natural frequencies considering the internal fluid ($f_{FF(tapered+SS+w)}$), and the simulation results of this paper. The low-depth model showed no significant difference in the error rate of the two natural frequency estimation equations, but the high-depth model had a large error.

Table 9. Comparison of natural frequency estimates for flexible soil (Equation (9)) without consideration of water (A), with consideration of water (B), and simulation result (C) ($\kappa = 1.0$).

Item	Water Depth	10 m	20 m	30 m	40 m	50 m	60 m	70 m
A	$f_{FF(tapered+SS)}$	0.2885	0.2662	0.2447	0.2241	0.2048	0.1866	0.1699
B	$f_{FF(tapered+SS+w)}$	0.2884	0.2658	0.2436	0.2221	0.2014	0.1820	0.1639
C	Results	0.2801	0.2558	0.2326	0.2106	0.1897	0.1701	0.1520
	Difference between A and C (%)	2.90	3.90	4.93	6.05	7.36	8.86	10.51
	Difference between B and C (%)	2.89	3.77	4.51	5.17	5.84	6.54	7.28

4.2. Comparison with Derived Equation for Considering Inner Water

In Section 4.2, a discussion of the impact of IWL is dealt with. By substituting Equations (6) and (7) into Equation (8), it can be expressed as Equation (20):

$$f_{RB(tapered+SS+w)} = \frac{1}{2\pi} \sqrt{\frac{k_{T(tapered+SS)}}{m_{RNA} + \alpha m_{T(tapered+SS)} + \psi \frac{\pi^2 \rho_w^2 H^3 (D_P - 2t_P)^4 \kappa^3}{32 m_{RNA} (L_T + H)}}} \quad (20)$$

where the ratio of water in the inner to the water depth (κ) means IWL with a range from 0 to 1. The empirical coefficient (ψ) created by the “inner fluid simplification assumption” can be assumed to be 0.0152 through the simulation results of this paper.

We can expect that the natural frequency will change non-linearly with a linear change of κ . Since all the variables constituting the term $\alpha_w m_w$ of Equation (8) are constants when the water depth, density of water, and the cross section of the substructure are constant, κ dominates. Figure 7 shows the natural frequency according to the change of IWL. The prediction that it would change non-linearly according to the change in κ was also found in the analysis results. Raising κ from 0.8 to 1.0 is about 3.7 times more effective than raising it from 0 to 0.2 in 10 m depth, while about 95.2 times more effective in 70m depth. Even if equal volumes of water are compared, it is clearly effective in deep water depth models.

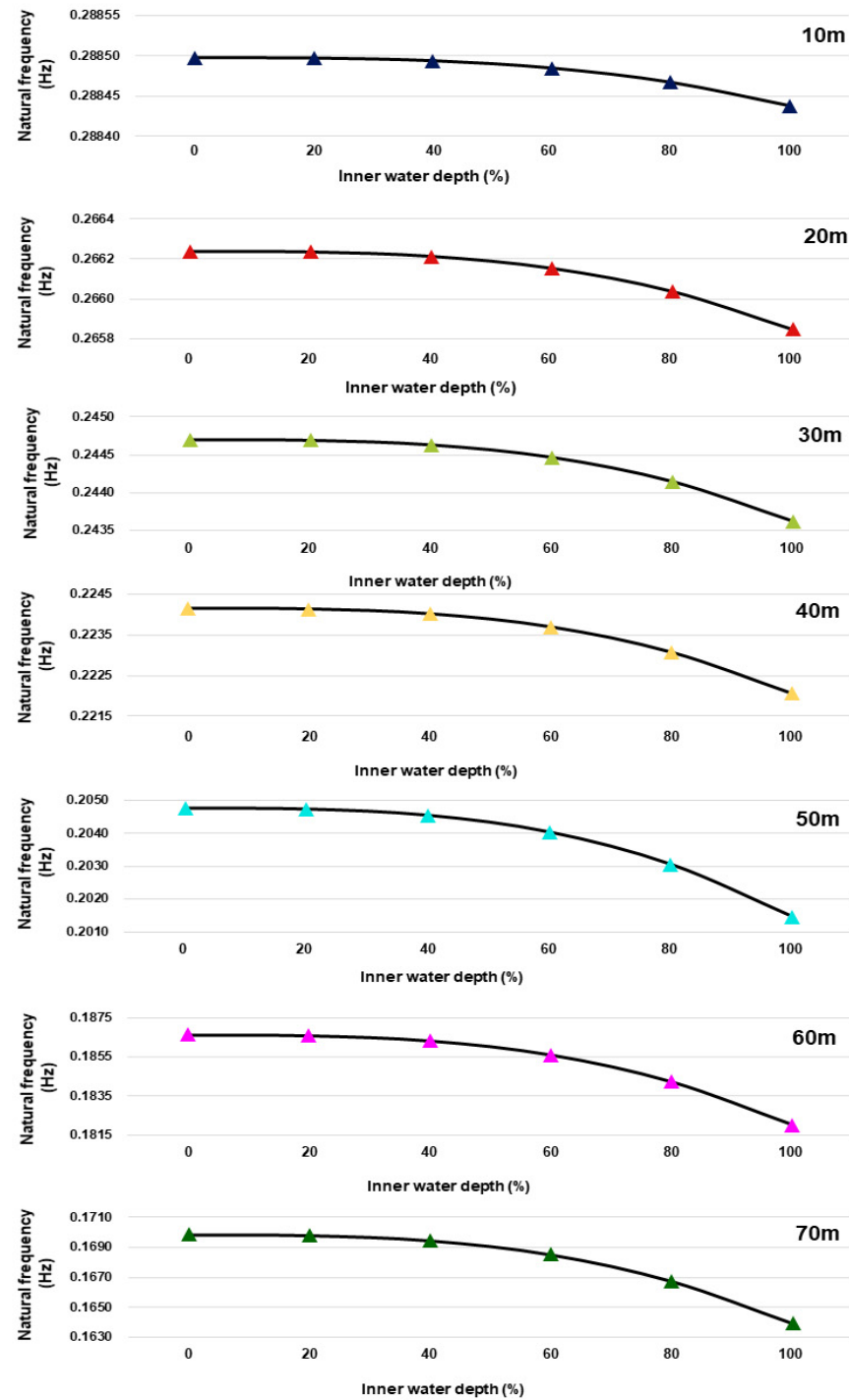


Figure 7. The natural frequency according to the change of IWL.

5. Conclusions

This paper is a study on the impact of natural frequency of interior fluid of fixed offshore wind turbine substructure and the development of a Variable Natural Frequency Damper (VNFD) for resonance avoidance of offshore wind turbine structures, and the conclusion of this study is as follows.

First, the inner water of fixed offshore wind turbine substructure affects the natural frequency of the wind turbine structure: the deeper the model, the larger the controllable natural frequency range. Specifically, the difference between fully supplied and drained

was very small at 0.027% in the 10 m water depth model, but was relatively large at 3.67% in the 70 m model.

Second, when estimating the natural frequency of fixed offshore wind turbine in deep water without consideration of interior fluid, a higher estimate is obtained than with consideration of it. The natural frequency estimation formula considering the internal fluid together with “Inner Fluid Simplification Assumption” had a smaller error rate when compared with the simulation than the estimation formula without considering the internal fluid. In particular, the difference in error rates increased rapidly as the water depth increased.

Third, even when controlling the same amount of water, controlling water at a high altitude has a greater effect on natural frequency. In particular, when 20% of the total was supplied from the lower part and upper part of the substructure, the difference in the effect on natural frequency was 3.7 times in the 10 m model and 95.2 times in the 70 m model.

As the water depth increases, the diameters of the monopile and the transition piece increase, and accordingly, the amount of fluid inside the substructure increases significantly. In this paper, the diameter of the substructure according to the water depth was kept constant. Therefore, in actual application, the natural frequency control effect through VNFD is expected to be greater. Also, the deeper the water depth, the higher the significant wave height, and this was considered in calculating the height of the maintenance platform when designing the offshore wind structure. In this study, the splash zone was not reflected in the FE model because water depth parameters were applied and wave height data were not used. Therefore, if the wave height is considered, the effect of VNFD is expected to be greater. In the deep water, for the application of this technique, the effect of hydrostatic pressure must be taken into account. The VNFD should be utilized so that the water level difference is not large and the controllable natural frequency is high.

A pressure-based formulation is used when considering fluid-structure interactions. Therefore, the tuning effect or sloshing mode of the inner water of the substructure was not analyzed with the “inner fluid simplification assumption”. However, as for other tuned dampers used in practice, it is expected that the inner water of the substructure will also have a damping effect due to the tuning effect. These simplifying assumptions should be simulated to reflect actual physical phenomena in future studies. In addition, the effect on other types of substructures will be different from those of monopile types, so this also requires further study.

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References

1. Ellingwood, B.R. Reliability-based condition assessment and LRFD for existing structures. *Struct. Saf.* **1996**, *18*, 67–80. [[CrossRef](#)]
2. *STANDARD DNV GL ST 0437*; Loads and Site Conditions for Wind Turbines. DNV GL: Oslo, Norway, 2016.
3. Sánchez, S.; López-Gutiérrez, J.-S.; Negro, V.; Esteban, M.D. Foundations in Offshore Wind Farms: Evolution, Characteristics and Range of Use. Analysis of Main Dimensional Parameters in Monopile Foundations. *J. Mar. Sci. Eng.* **2019**, *7*, 441. [[CrossRef](#)]

4. Europe, W. Offshore Wind in Europe: Key Trends and Statistics. 2018. Available online: <https://windeurope.org/about-wind/statistics/offshore/european-offshore-wind-industry-key-trends-statistics-2018/> (accessed on 14 May 2022).
5. DNVGL-ST-0126; Support Structures for Wind Turbines. DNV GL: Oslo, Norway, 2016.
6. Devriendt, C.; Weijtjens, W.; El-Kafafy, M.; De Sitter, G. Monitoring resonant frequencies and damping values of an offshore wind turbine in parked conditions. *IET Renew. Power Gener.* **2014**, *8*, 433–441. [\[CrossRef\]](#)
7. Sunday, K.; Brennan, F. A review of offshore wind monopiles structural design achievements and challenges. *Ocean Eng.* **2021**, *235*, 109409. [\[CrossRef\]](#)
8. Arany, L.; Bhattacharya, S.; Macdonald, J.H.; Hogan, S.J. Closed form solution of Eigen frequency of monopile supported offshore wind turbines in deeper waters incorporating stiffness of substructure and SSI. *Soil Dyn. Earthq. Eng.* **2016**, *83*, 18–32. [\[CrossRef\]](#)
9. Hilbert, L.; Black, A.; Andersen, F.; Mathiesen, T. Inspection and monitoring of corrosion inside monopile foundations for offshore wind turbines. *Eur. Corros. Congr.* **2011**, *3*, 2187–2201.
10. Damgaard, M.; Andersen, J.K. Natural frequency and damping estimation of an offshore wind turbine structure. In Proceedings of the the Twenty-Second International Offshore and Polar Engineering Conference, Rhodes, Greece, 17–22 June 2012.
11. Kallehave, D.; Thilsted, C.L.; Liingaard, M. Modification of the API py formulation of initial stiffness of sand. In Proceedings of the Offshore Site Investigation and Geotechnics: Integrated Technologies-Present and Future, London, UK, 12–14 September 2012.
12. Kühn, M. Soft or stiff-a fundamental question in the design of offshore wind energy converters. In Proceedings of the European Wind Energy Conference, Dublin Castle, Ireland, 6–9 October 1997; pp. 575–578.
13. Lowe, J. Hornsea Met Mast-A Demonstration of the ‘Twisted Jacket’ Design. In Proceedings of the Future Offshore Wind Turbine Foundation Conference, Bremen, Germany, 2 February 2012.
14. Zaaier, M. *Design Methods for Offshore Wind Turbines at Exposed Sites (OWTES)—Sensitivity Analysis for Foundations of Offshore Wind Turbines*; Garrad Hassan: Delft, The Netherlands, 2002.
15. Zania, V. Natural vibration frequency and damping of slender structures founded on monopiles. *Soil Dyn. Earthq. Eng.* **2014**, *59*, 8–20. [\[CrossRef\]](#)
16. Prendergast, L.J.; Gavin, K.; Doherty, P. An investigation into the effect of scour on the natural frequency of an offshore wind turbine. *Ocean Eng.* **2015**, *101*, 1–11. [\[CrossRef\]](#)
17. Yi, J.-H.; Kim, S.-B.; Yoon, G.-L.; Andersen, L.V. Natural frequency of bottom-fixed offshore wind turbines considering pile-soil-interaction with material uncertainties and scouring depth. *Wind. Struct.* **2015**, *21*, 625–639. [\[CrossRef\]](#)
18. van der Tempel, J.; Molenaar, D.-P. Wind turbine structural dynamics—A review of the principles for modern power generation, onshore and offshore. *Wind Eng.* **2002**, *26*, 211–222. [\[CrossRef\]](#)
19. Adhikari, S.; Bhattacharya, S. Dynamic analysis of wind turbine towers on flexible foundations. *Shock Vib.* **2012**, *19*, 37–56. [\[CrossRef\]](#)
20. Ko, Y.-Y. A simplified structural model for monopile-supported offshore wind turbines with tapered towers. *Renew. Energy* **2020**, *156*, 777–790. [\[CrossRef\]](#)
21. Wang, P.; Xu, Y.; Zhang, X.; Xi, R.; Du, X. A substructure method for seismic responses of offshore wind turbine considering nonlinear pile-soil dynamic interaction. *Soil Dyn. Earthq. Eng.* **2021**, *144*, 106684. [\[CrossRef\]](#)
22. Darvishi-Alamouti, S.; Bahaari, M.-R.; Moradi, M. Natural frequency of offshore wind turbines on rigid and flexible monopiles in cohesionless soils with linear stiffness distribution. *Appl. Ocean. Res.* **2017**, *68*, 91–102. [\[CrossRef\]](#)
23. Kramer, S.L. *Geotechnical Earthquake Engineering*; Pearson Education India: Nodia, India, 1996.
24. Jhung, M.J.; Kang, S.-S. Fluid effect on the modal characteristics of a square tank. *Nucl. Eng. Technol.* **2019**, *51*, 1117–1131. [\[CrossRef\]](#)
25. Morand, H.J.-P.; Ohayon, R. *Fluid Structure Interaction*. Wiley & Sons. 1995. Available online: <https://onlinelibrary.wiley.com/doi/10.1002/zamm.19960760705> (accessed on 14 May 2022).
26. Benra, F.-K.; Dohmen, H.J.; Pei, J.; Schuster, S.; Wan, B. A comparison of one-way and two-way coupling methods for numerical analysis of fluid-structure interactions. *J. Appl. Math.* **2011**, *2011*, 853560. [\[CrossRef\]](#)
27. Gutknecht, M.H. The unsymmetric Lanczos algorithms and their relations to Pade approximation, continued fractions and the QD algorithm. In Proceedings of the Copper Mountain Conference on Iterative Methods, Copper Mountain, CO, USA, 9–13 April 1990.
28. Rajakumar, C.; Rogers, C. The Lanczos algorithm applied to unsymmetric generalized eigenvalue problem. *Int. J. Numer. Methods Eng.* **1991**, *32*, 1009–1026. [\[CrossRef\]](#)
29. Bankar, S.S. *Vibration and Acoustic Analysis of Laminated Composite Plate*. Ph.D. Thesis, National Institute of Technology Rourkela, Odisha, India, May 2015.
30. American Petroleum Institute. *Planning, Designing, and Constructing Fixed Offshore Platforms—Working Stress Design*; American Petroleum Institute: Washington, DC, USA, 2014.
31. DNVGL-RP-C212; Offshore Soil Mechanics and Geotechnical Engineering. DNV GL: Høvik, Norway, 2017.
32. Abdullahi, A.; Wang, Y.; Bhattacharya, S. Comparative Modal Analysis of Monopile and Jacket Supported Offshore Wind Turbines including Soil-Structure Interaction. *Int. J. Struct. Stab. Dyn.* **2020**, *20*, 2042016. [\[CrossRef\]](#)
33. Jalbi, S.; Shadlou, M.; Bhattacharya, S. Practical method to estimate foundation stiffness for design of offshore wind turbines. In *Wind Energy Engineering*; Elsevier: Amsterdam, The Netherlands, 2017; pp. 329–352.

34. Abhinav, K.; Saha, N. Dynamic analysis of an offshore wind turbine including soil effects. *Procedia Eng.* **2015**, *116*, 32–39. [[CrossRef](#)]
35. Jonkman, J.; Butterfield, S.; Musial, W.; Scott, G. *Definition of a 5-MW Reference Wind Turbine for Offshore System Development*; National Renewable Energy Lab. (NREL): Golden, CO, USA, 2009.
36. Gentils, T.; Wang, L.; Kolios, A. Integrated structural optimisation of offshore wind turbine support structures based on finite element analysis and genetic algorithm. *Appl. Energy* **2017**, *199*, 187–204. [[CrossRef](#)]
37. Shi, W.; Park, H.-C.; Baek, J.-H.; Kim, C.-W.; Kim, Y.-C.; Shin, H.-K. Study on the marine growth effect on the dynamic response of offshore wind turbines. *Int. J. Precis. Eng. Manuf.* **2012**, *13*, 1167–1176. [[CrossRef](#)]
38. Jung, S.; Kim, S.-R.; Patil, A. Effect of monopile foundation modeling on the structural response of a 5-MW offshore wind turbine tower. *Ocean Eng.* **2015**, *109*, 479–488. [[CrossRef](#)]
39. ANSYS Inc. Contact Technology Guide Release 12.1. 2009. Available online: <https://www.cae.tntech.edu> (accessed on 14 May 2022).
40. Özer, H.; Öz, Ö. Three dimensional finite element analysis of bi-adhesively bonded double lap joint. *Int. J. Adhes. Adhes.* **2012**, *37*, 50–55. [[CrossRef](#)]
41. Lanoue, F.; Vadean, A.; Sanschagrin, B. Finite element analysis and contact modelling considerations of interference fits for fretting fatigue strength calculations. *Simul. Model. Pract. Theory* **2009**, *17*, 1587–1602. [[CrossRef](#)]
42. Daneshi, M. Numerical investigation of the fluid flow around and past a circular cylinder by ANSYS simulation. *Int. J. Adv. Sci. Technol.* **2016**, *92*, 49–58. [[CrossRef](#)]
43. Tran, T.; Kim, D.; Song, J. Computational fluid dynamic analysis of a floating offshore wind turbine experiencing platform pitching motion. *Energies* **2014**, *7*, 5011–5026. [[CrossRef](#)]
44. Kolios, A.; Wang, L.; Mehmanparast, A.; Brennan, F. Determination of stress concentration factors in offshore wind welded structures through a hybrid experimental and numerical approach. *Ocean Eng.* **2019**, *178*, 38–47. [[CrossRef](#)]
45. Jonkman, J.; Musial, W. *Offshore Code Comparison Collaboration (OC3) for IEA Wind Task 23 Offshore Wind Technology and Deployment*; National Renewable Energy Lab. (NREL): Golden, CO, USA, 2010.
46. Carswell, W. Probabilistic Analysis of Offshore Wind Turbine Soil-Structure Interaction. Master's Thesis, University of Massachusetts Amherst, Amherst, MA, USA, 2012.